Human effluent treatment project report

Hungary - November 2012

During November 2012 a feasibility study was made at Bonyhad municipal effluent treatment plant to measure the Dissolved Oxygen (DO) increasing efficiency of Vortex Flowform and the anaerobe smell reducing capacity on human effluent.

Two type of test were carried out, one using a 7 unit Vortex Flowform system treating the primary human effluent of 460m3, the other one using a 3 unit Vortex system to treat 4m3 primary effluent stored in a 5m3 tank.

The 460m3 volume primary effluent is kept in an oval open air pool (picture 1) and being circulated at a speed of 40cm/sec before being treated. No aeration of any kind is performed thus the resulting smell can reach the town at favorable wind direction and intensity.

Picture 1

The 7 unit Vortex was installed at the side of the pool, see Picture 2. An underwater pump was feeding the system, while the treated effluent flowing directly back to the main pool, see Picture 3.

Picture 2

Picture 3

Level of DO was measured in the pool before the Vortex system and straight after, as well as in each of the seven forms. Temperature of the effluent during measurements was 14,1C.

Results can be seen in the below table:

Location	Before Vortex	After Vortex	Vortex 1	Vortex 2	Vortex 3	Vortex 4	Vortex 5	Vortex 6	Vortex 7
Level of DO mg/l	**0,26**	**0,39**	*1,15*	*1,67*	*2,67*	*3,60*	*4,56*	*5,67*	*7,01*

Table 1

While the amount of DO has increased approx. 1 mg/lit in each Vortex unit, the impact of the treated amount (approx. 200 lit/min) on the total effluent pool can only be measured straight after the Vortex: 0,39 mg/lit vs. 0,26 mg/lit. The level of DO in the pool a few meters after the Vortex was back to 0,26 mg/l, same as before the Vortex Flowform.

The flow rate of the Vortex system being approx. 200 lit/min, the total amount of effluent is treated once in every 38 hours. In spite that every day approx. 70m3 fresh effluent was discharged in the pool the level of **average DO has increased from 0,2 to 0,4 mg/lit** over a period of a few weeks due to continuous operation of the Vortex system.

In order to eliminate smell, an aerobe process is required in the top 20-30% layer of the entire effluent pool. By experience of staff from the local effluent treatment plant a min 2 mg/lit DO level is necessary in this top layer.
Calculation method: Isolate certain amount of effluent (4m3 in our case, see below) and treat non-stop with Vortex to enable maximum oxygen input, while measuring the Chemical Oxygen Demand (COD) of the effluent on daily basis. In our case this was approx. 40 g/m3/day (120 mg/lit difference in 3 days) that is 1,67 g/m3/h. One Vortex Oxygen input capacity is approx. 1 mg/lit (Table 1) at 12 m3/h speed, which is 12 g/h. Thus one Vortex unit can treat approx. 7 m3 (12/1,7) at this speed. To treat the 20% of the 460m3 = 92m3 minimum 13 Vortex units are required, while for the top 30% of the effluent volume (138m3) minimum 20 Vortex units have to be installed. This is true in case the Vortex treated high DO content effluent can be evenly spread across the whole surface on continuous basis. Temperature can also impact the efficiency.

The project has however not been completed with this increased amount of Flowform due to lack of funds to support manufacturing and installation.

For calculating the efficiency, another project with an isolated 4 m3 primary untreated human effluent was carried out while measuring more parameters. The effluent was stored in a 5m3 tank; see Picture 4 and being constantly circulated through a set of 3 Vortex units for a period of 14 days. Several parameters were measured; results can be seen in Table 2.

Picture 4

Parameters / Dates	9 Nov	12 Nov	13 Nov	14 Nov	15 Nov	16 Nov	21 Nov	23 Nov
Temp °C	12,5	14,7	12,8	12,8	11,6	10,7	12,8	13,0
DO mg/lit	0,34	6,58	9,30	9,30	9,95	10,72	10,72	10,72
DO stand 25°C	0,26	5,27	7,33	7,33	7,55	7,98	8,36	8,40
ph	8,02	8,37	8,19	8,85	8,43	8,87	8,54	8,71
COD mg O2/lit	480	360	535	400	315	340	200	220
BOD mg O2/lit	380			316		330	84	29
NH4 mg/lit	79,86	69,85	70,05	67,25	61,64	58,84	54,64	56,04
NO2 -	0,00	1,00	1,00	10	30	20	40	60
NO3 -	0,00	25	10	100	200	250	250	500
Total N2 mg/lit	217,1			146,4		126,8	98,3	117,4

Table 2

Measurement results reveal the significant increase of DO during first 3 days of operation and start of the aerobic and nitrification process. Both Chemical Oxygen Demand (COD) and Biological Oxygen Demand (BOD) have significantly reduced. Smell could not be observed after 2nd day of operation and the black effluent became semitransparent odorless water.